

FarmConnect Database Schema (PostgreSQL)

```
-- Migration: 20260310133950_init_phase2
-- CreateEnum
CREATE TYPE "Role" AS ENUM ('PRODUCER', 'BUYER', 'ADMIN');

-- CreateEnum
CREATE TYPE "UserStatus" AS ENUM ('ACTIVE', 'SUSPENDED', 'PENDING_VERIFICATION', 'DELETED');

-- CreateEnum
CREATE TYPE "VerificationStatus" AS ENUM ('UNVERIFIED', 'PENDING', 'APPROVED', 'REJECTED');

-- CreateEnum
CREATE TYPE "ProductStatus" AS ENUM ('DRAFT', 'ACTIVE', 'INACTIVE', 'OUT_OF_STOCK', 'ARCHIVED');

-- CreateEnum
CREATE TYPE "OrderStatus" AS ENUM ('PENDING', 'CONFIRMED', 'PROCESSING', 'SHIPPED', 'DELIVERED', 'CANCELLED', 'REFUNDED');

-- CreateEnum
CREATE TYPE "PaymentStatus" AS ENUM ('UNPAID', 'PENDING', 'PAID', 'FAILED', 'REFUNDED', 'PARTIALLY_REFUNDED');

-- CreateEnum
CREATE TYPE "DeliveryStatus" AS ENUM ('NOT_SHIPPED', 'PREPARING', 'IN_TRANSIT', 'OUT_FOR_DELIVERY', 'DELIVERED', 'FAILED_DELIVERY', 'RETURNED');

-- CreateTable
CREATE TABLE "users" (
    "id" TEXT NOT NULL,
    "email" TEXT NOT NULL,
    "passwordHash" TEXT NOT NULL,
    "fullName" TEXT NOT NULL,
    "role" "Role" NOT NULL DEFAULT 'BUYER',
    "status" "UserStatus" NOT NULL DEFAULT 'PENDING_VERIFICATION',
    "avatarUrl" TEXT,
    "phone" TEXT,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,
    "deletedAt" TIMESTAMP(3),

    CONSTRAINT "users_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "refresh_tokens" (
    "id" TEXT NOT NULL,
    "token" TEXT NOT NULL,
    "userId" TEXT NOT NULL,
    "expiresAt" TIMESTAMP(3) NOT NULL,
    "revokedAt" TIMESTAMP(3),
    "userAgent" TEXT,
    "ipAddress" TEXT,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "refresh_tokens_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "producer_profiles" (
    "id" TEXT NOT NULL,
    "userId" TEXT NOT NULL,
    "businessName" TEXT NOT NULL,
    "businessType" TEXT,
    "bio" TEXT,
    "wilaya" TEXT NOT NULL,
    "commune" TEXT NOT NULL,
    "nif" TEXT,
    "nifDocumentUrl" TEXT,
    "verificationStatus" "VerificationStatus" NOT NULL DEFAULT 'UNVERIFIED',
    "verifiedAt" TIMESTAMP(3),
    "verifiedById" TEXT,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "producer_profiles_pkey" PRIMARY KEY ("id")
);
```

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```
-- CreateTable
CREATE TABLE "producer_verification_requests" (
  "id" TEXT NOT NULL,
  "producerProfileId" TEXT NOT NULL,
  "status" "VerificationStatus" NOT NULL DEFAULT 'PENDING',
  "notes" TEXT,
  "documents" JSONB NOT NULL DEFAULT '[]',
  "submittedAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "reviewedAt" TIMESTAMP(3),
  "reviewedById" TEXT,

  CONSTRAINT "producer_verification_requests_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "buyer_profiles" (
  "id" TEXT NOT NULL,
  "userId" TEXT NOT NULL,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "updatedAt" TIMESTAMP(3) NOT NULL,

  CONSTRAINT "buyer_profiles_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "addresses" (
  "id" TEXT NOT NULL,
  "userId" TEXT NOT NULL,
  "label" TEXT NOT NULL DEFAULT 'home',
  "recipientName" TEXT NOT NULL,
  "phone" TEXT NOT NULL,
  "wilaya" TEXT NOT NULL,
  "commune" TEXT NOT NULL,
  "street" TEXT NOT NULL,
  "postalCode" TEXT,
  "isDefault" BOOLEAN NOT NULL DEFAULT false,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "updatedAt" TIMESTAMP(3) NOT NULL,

  CONSTRAINT "addresses_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "categories" (
  "id" TEXT NOT NULL,
  "name" JSONB NOT NULL,
  "slug" TEXT NOT NULL,
  "parentId" TEXT,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "updatedAt" TIMESTAMP(3) NOT NULL,

  CONSTRAINT "categories_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "products" (
  "id" TEXT NOT NULL,
  "producerId" TEXT NOT NULL,
  "categoryId" TEXT,
  "title" JSONB NOT NULL,
  "description" JSONB NOT NULL,
  "slug" TEXT NOT NULL,
  "price" DECIMAL(10,2) NOT NULL,
  "currency" TEXT NOT NULL DEFAULT 'DZD',
  "unit" TEXT NOT NULL DEFAULT 'kg',
  "stock" INTEGER NOT NULL DEFAULT 0,
  "minOrderQty" INTEGER NOT NULL DEFAULT 1,
  "maxOrderQty" INTEGER NOT NULL DEFAULT 10000,
  "status" "ProductStatus" NOT NULL DEFAULT 'DRAFT',
  "tags" TEXT[] DEFAULT ARRAY[]::TEXT[],
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  "updatedAt" TIMESTAMP(3) NOT NULL,
  "deletedAt" TIMESTAMP(3),
```

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```
CONSTRAINT "products_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "product_images" (
    "id" TEXT NOT NULL,
    "productId" TEXT NOT NULL,
    "url" TEXT NOT NULL,
    "altText" TEXT,
    "position" INTEGER NOT NULL DEFAULT 0,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "product_images_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "carts" (
    "id" TEXT NOT NULL,
    "buyerId" TEXT NOT NULL,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "carts_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "cart_items" (
    "id" TEXT NOT NULL,
    "cartId" TEXT NOT NULL,
    "productId" TEXT NOT NULL,
    "quantity" INTEGER NOT NULL,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "cart_items_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "orders" (
    "id" TEXT NOT NULL,
    "buyerId" TEXT NOT NULL,
    "buyerAddressId" TEXT,
    "status" "OrderStatus" NOT NULL DEFAULT 'PENDING',
    "paymentStatus" "PaymentStatus" NOT NULL DEFAULT 'UNPAID',
    "deliveryStatus" "DeliveryStatus" NOT NULL DEFAULT 'NOT_SHIPPED',
    "total" DECIMAL(10,2) NOT NULL,
    "currency" TEXT NOT NULL DEFAULT 'DZD',
    "notes" TEXT,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "orders_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "order_items" (
    "id" TEXT NOT NULL,
    "orderId" TEXT NOT NULL,
    "productId" TEXT,
    "productSnapshot" JSONB NOT NULL,
    "quantity" INTEGER NOT NULL,
    "unitPrice" DECIMAL(10,2) NOT NULL,
    "total" DECIMAL(10,2) NOT NULL,
    "currency" TEXT NOT NULL DEFAULT 'DZD',

    CONSTRAINT "order_items_pkey" PRIMARY KEY ("id")
);

-- CreateIndex
CREATE UNIQUE INDEX "users_email_key" ON "users"("email");

-- CreateIndex
CREATE INDEX "users_role_status_idx" ON "users"("role", "status");

-- CreateIndex
```

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```
CREATE UNIQUE INDEX "refresh_tokens_token_key" ON "refresh_tokens"("token");

-- CreateIndex
CREATE INDEX "refresh_tokens_userId_idx" ON "refresh_tokens"("userId");

-- CreateIndex
CREATE UNIQUE INDEX "producer_profiles_userId_key" ON "producer_profiles"("userId");

-- CreateIndex
CREATE UNIQUE INDEX "producer_profiles_nif_key" ON "producer_profiles"("nif");

-- CreateIndex
CREATE INDEX "producer_profiles_wilaya_idx" ON "producer_profiles"("wilaya");

-- CreateIndex
CREATE INDEX "producer_profiles_verificationStatus_idx" ON "producer_profiles"("verificationStatus");

-- CreateIndex
CREATE INDEX "producer_verification_requests_producerProfileId_idx" ON
  "producer_verification_requests"("producerProfileId");

-- CreateIndex
CREATE INDEX "producer_verification_requests_status_idx" ON "producer_verification_requests"("status");

-- CreateIndex
CREATE UNIQUE INDEX "buyer_profiles_userId_key" ON "buyer_profiles"("userId");

-- CreateIndex
CREATE INDEX "addresses_userId_idx" ON "addresses"("userId");

-- CreateIndex
CREATE UNIQUE INDEX "categories_slug_key" ON "categories"("slug");

-- CreateIndex
CREATE UNIQUE INDEX "products_slug_key" ON "products"("slug");

-- CreateIndex
CREATE INDEX "products_producerId_idx" ON "products"("producerId");

-- CreateIndex
CREATE INDEX "products_categoryId_idx" ON "products"("categoryId");

-- CreateIndex
CREATE INDEX "products_status_idx" ON "products"("status");

-- CreateIndex
CREATE INDEX "products_slug_idx" ON "products"("slug");

-- CreateIndex
CREATE INDEX "product_images_productId_idx" ON "product_images"("productId");

-- CreateIndex
CREATE UNIQUE INDEX "carts_buyerId_key" ON "carts"("buyerId");

-- CreateIndex
CREATE INDEX "cart_items_cartId_idx" ON "cart_items"("cartId");

-- CreateIndex
CREATE UNIQUE INDEX "cart_items_cartId_productId_key" ON "cart_items"("cartId", "productId");

-- CreateIndex
CREATE INDEX "orders_buyerId_idx" ON "orders"("buyerId");

-- CreateIndex
CREATE INDEX "orders_status_idx" ON "orders"("status");

-- CreateIndex
CREATE INDEX "orders_paymentStatus_idx" ON "orders"("paymentStatus");

-- CreateIndex
CREATE INDEX "order_items_orderId_idx" ON "order_items"("orderId");

-- AddForeignKey
ALTER TABLE "refresh_tokens" ADD CONSTRAINT "refresh_tokens_userId_fkey" FOREIGN KEY ("userId") REFERENCES
  "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;
```

FarmConnect Database Schema (PostgreSQL)

```
-- AddForeignKey
ALTER TABLE "producer_profiles" ADD CONSTRAINT "producer_profiles_userId_fkey" FOREIGN KEY ("userId")
REFERENCES "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "producer_verification_requests" ADD CONSTRAINT
"producer_verification_requests_producerProfileId_fkey" FOREIGN KEY ("producerProfileId") REFERENCES
"producer_profiles"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "producer_verification_requests" ADD CONSTRAINT
"producer_verification_requests_reviewedById_fkey" FOREIGN KEY ("reviewedById") REFERENCES "users"("id") ON
DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "buyer_profiles" ADD CONSTRAINT "buyer_profiles_userId_fkey" FOREIGN KEY ("userId") REFERENCES
"users"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "addresses" ADD CONSTRAINT "addresses_userId_fkey" FOREIGN KEY ("userId") REFERENCES
"users"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "categories" ADD CONSTRAINT "categories_parentId_fkey" FOREIGN KEY ("parentId") REFERENCES
"categories"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "products" ADD CONSTRAINT "products_producerId_fkey" FOREIGN KEY ("producerId") REFERENCES
"producer_profiles"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "products" ADD CONSTRAINT "products_categoryId_fkey" FOREIGN KEY ("categoryId") REFERENCES
"categories"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "product_images" ADD CONSTRAINT "product_images_productId_fkey" FOREIGN KEY ("productId")
REFERENCES "products"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "carts" ADD CONSTRAINT "carts_buyerId_fkey" FOREIGN KEY ("buyerId") REFERENCES "users"("id") ON
DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "cart_items" ADD CONSTRAINT "cart_items_cartId_fkey" FOREIGN KEY ("cartId") REFERENCES
"carts"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "cart_items" ADD CONSTRAINT "cart_items_productId_fkey" FOREIGN KEY ("productId") REFERENCES
"products"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "orders" ADD CONSTRAINT "orders_buyerId_fkey" FOREIGN KEY ("buyerId") REFERENCES "users"("id") ON
DELETE RESTRICT ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "orders" ADD CONSTRAINT "orders_buyerAddressId_fkey" FOREIGN KEY ("buyerAddressId") REFERENCES
"addresses"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "order_items" ADD CONSTRAINT "order_items_orderId_fkey" FOREIGN KEY ("orderId") REFERENCES
"orders"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "order_items" ADD CONSTRAINT "order_items_productId_fkey" FOREIGN KEY ("productId") REFERENCES
"products"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- Migration: 20260310135305_phase2c
-- CreateEnum
CREATE TYPE "PaymentMethod" AS ENUM ('CASH_ON_DELIVERY', 'BANK_TRANSFER', 'CIB_CARD', 'EDAHABIA');

-- CreateEnum
CREATE TYPE "PaymentEventType" AS ENUM ('INITIATED', 'AUTHORIZED', 'CAPTURED', 'FAILED', 'REFUNDED',
'CHARGEBACK');
```

FarmConnect Database Schema (PostgreSQL)

```
-- CreateEnum
CREATE TYPE "PayoutStatus" AS ENUM ('PENDING', 'PROCESSING', 'PAID', 'FAILED');

-- CreateEnum
CREATE TYPE "NotificationType" AS ENUM ('ORDER_PLACED', 'ORDER_CONFIRMED', 'ORDER_SHIPPED',
    'ORDER_DELIVERED', 'ORDER_CANCELLED', 'PAYMENT_SUCCESS', 'PAYMENT_FAILED', 'PRODUCT_LOW_STOCK',
    'PRODUCER_VERIFIED', 'PRODUCER_REJECTED', 'GENERAL');

-- CreateEnum
CREATE TYPE "NotificationChannel" AS ENUM ('IN_APP', 'EMAIL', 'SMS', 'PUSH');

-- CreateEnum
CREATE TYPE "AuditAction" AS ENUM ('CREATE', 'UPDATE', 'DELETE', 'LOGIN', 'LOGOUT', 'VERIFY', 'REJECT',
    'SUSPEND', 'UNSUSPEND');

-- CreateTable
CREATE TABLE "payments" (
    "id" TEXT NOT NULL,
    "orderId" TEXT NOT NULL,
    "method" "PaymentMethod" NOT NULL,
    "status" "PaymentStatus" NOT NULL DEFAULT 'PENDING',
    "amount" DECIMAL(10,2) NOT NULL,
    "currency" TEXT NOT NULL DEFAULT 'DZD',
    "gatewayRef" TEXT,
    "gatewayResponse" JSONB,
    "paidAt" TIMESTAMP(3),
    "failedAt" TIMESTAMP(3),
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "payments_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "payment_events" (
    "id" TEXT NOT NULL,
    "paymentId" TEXT NOT NULL,
    "type" "PaymentEventType" NOT NULL,
    "payload" JSONB NOT NULL DEFAULT '{}',
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "payment_events_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "payouts" (
    "id" TEXT NOT NULL,
    "producerProfileId" TEXT NOT NULL,
    "amount" DECIMAL(10,2) NOT NULL,
    "currency" TEXT NOT NULL DEFAULT 'DZD',
    "status" "PayoutStatus" NOT NULL DEFAULT 'PENDING',
    "bankName" TEXT,
    "bankAccount" TEXT,
    "reference" TEXT,
    "notes" TEXT,
    "processedAt" TIMESTAMP(3),
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "payouts_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "delivery_trackings" (
    "id" TEXT NOT NULL,
    "orderId" TEXT NOT NULL,
    "status" "DeliveryStatus" NOT NULL,
    "location" TEXT,
    "description" TEXT,
    "occurredAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "delivery_trackings_pkey" PRIMARY KEY ("id")
);
```

FarmConnect Database Schema (PostgreSQL)

```
-- CreateTable
CREATE TABLE "notifications" (
    "id" TEXT NOT NULL,
    "userId" TEXT NOT NULL,
    "type" "NotificationType" NOT NULL,
    "channel" "NotificationChannel" NOT NULL DEFAULT 'IN_APP',
    "title" TEXT NOT NULL,
    "body" TEXT NOT NULL,
    "data" JSONB NOT NULL DEFAULT '{}',
    "readAt" TIMESTAMP(3),
    "sentAt" TIMESTAMP(3),
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "notifications_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "analytics_daily" (
    "id" TEXT NOT NULL,
    "date" DATE NOT NULL,
    "wilaya" TEXT,
    "newUsers" INTEGER NOT NULL DEFAULT 0,
    "newProducers" INTEGER NOT NULL DEFAULT 0,
    "newBuyers" INTEGER NOT NULL DEFAULT 0,
    "totalOrders" INTEGER NOT NULL DEFAULT 0,
    "totalRevenue" DECIMAL(14,2) NOT NULL DEFAULT 0,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
    "updatedAt" TIMESTAMP(3) NOT NULL,

    CONSTRAINT "analytics_daily_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "ai_recommendations" (
    "id" TEXT NOT NULL,
    "userId" TEXT NOT NULL,
    "productId" TEXT NOT NULL,
    "score" DOUBLE PRECISION NOT NULL,
    "reason" TEXT,
    "seenAt" TIMESTAMP(3),
    "clickedAt" TIMESTAMP(3),
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "ai_recommendations_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "ai_forecasts" (
    "id" TEXT NOT NULL,
    "productId" TEXT,
    "categoryId" TEXT,
    "forecastDate" DATE NOT NULL,
    "predictedDemand" DECIMAL(12,2) NOT NULL,
    "confidenceScore" DOUBLE PRECISION NOT NULL,
    "modelVersion" TEXT NOT NULL DEFAULT 'v1',
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "ai_forecasts_pkey" PRIMARY KEY ("id")
);

-- CreateTable
CREATE TABLE "audit_logs" (
    "id" TEXT NOT NULL,
    "actorId" TEXT,
    "targetType" TEXT NOT NULL,
    "targetId" TEXT NOT NULL,
    "action" "AuditAction" NOT NULL,
    "changes" JSONB NOT NULL DEFAULT '{}',
    "ipAddress" TEXT,
    "userAgent" TEXT,
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT "audit_logs_pkey" PRIMARY KEY ("id")
);
```

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```
-- CreateIndex
CREATE UNIQUE INDEX "payments_orderId_key" ON "payments"("orderId");

-- CreateIndex
CREATE INDEX "payments_status_idx" ON "payments"("status");

-- CreateIndex
CREATE INDEX "payments_method_idx" ON "payments"("method");

-- CreateIndex
CREATE INDEX "payment_events_paymentId_idx" ON "payment_events"("paymentId");

-- CreateIndex
CREATE INDEX "payouts_producerProfileId_idx" ON "payouts"("producerProfileId");

-- CreateIndex
CREATE INDEX "payouts_status_idx" ON "payouts"("status");

-- CreateIndex
CREATE INDEX "delivery_trackings_orderId_idx" ON "delivery_trackings"("orderId");

-- CreateIndex
CREATE INDEX "notifications_userId_readAt_idx" ON "notifications"("userId", "readAt");

-- CreateIndex
CREATE INDEX "notifications_userId_createdAt_idx" ON "notifications"("userId", "createdAt");

-- CreateIndex
CREATE INDEX "analytics_daily_date_idx" ON "analytics_daily"("date");

-- CreateIndex
CREATE INDEX "analytics_daily_wilaya_idx" ON "analytics_daily"("wilaya");

-- CreateIndex
CREATE INDEX "ai_recommendations_userId_score_idx" ON "ai_recommendations"("userId", "score");

-- CreateIndex
CREATE UNIQUE INDEX "ai_recommendations_userId_productId_key" ON "ai_recommendations"("userId", "productId");

-- CreateIndex
CREATE INDEX "ai_forecasts_productId_forecastDate_idx" ON "ai_forecasts"("productId", "forecastDate");

-- CreateIndex
CREATE INDEX "ai_forecasts_categoryId_forecastDate_idx" ON "ai_forecasts"("categoryId", "forecastDate");

-- CreateIndex
CREATE INDEX "audit_logs_actorId_idx" ON "audit_logs"("actorId");

-- CreateIndex
CREATE INDEX "audit_logs_targetType_targetId_idx" ON "audit_logs"("targetType", "targetId");

-- CreateIndex
CREATE INDEX "audit_logs_createdAt_idx" ON "audit_logs"("createdAt");

-- AddForeignKey
ALTER TABLE "payments" ADD CONSTRAINT "payments_orderId_fkey" FOREIGN KEY ("orderId") REFERENCES
  "orders"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "payment_events" ADD CONSTRAINT "payment_events_paymentId_fkey" FOREIGN KEY ("paymentId")
  REFERENCES "payments"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "payouts" ADD CONSTRAINT "payouts_producerProfileId_fkey" FOREIGN KEY ("producerProfileId")
  REFERENCES "producer_profiles"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "delivery_trackings" ADD CONSTRAINT "delivery_trackings_orderId_fkey" FOREIGN KEY ("orderId")
  REFERENCES "orders"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "notifications" ADD CONSTRAINT "notifications_userId_fkey" FOREIGN KEY ("userId") REFERENCES
  "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
```


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```
ALTER TABLE "ai_recommendations" ADD CONSTRAINT "ai_recommendations_userId_fkey" FOREIGN KEY ("userId")
REFERENCES "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "ai_recommendations" ADD CONSTRAINT "ai_recommendations_productId_fkey" FOREIGN KEY ("productId")
REFERENCES "products"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "ai_forecasts" ADD CONSTRAINT "ai_forecasts_productId_fkey" FOREIGN KEY ("productId") REFERENCES
"products"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "ai_forecasts" ADD CONSTRAINT "ai_forecasts_categoryId_fkey" FOREIGN KEY ("categoryId")
REFERENCES "categories"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- AddForeignKey
ALTER TABLE "audit_logs" ADD CONSTRAINT "audit_logs_actorId_fkey" FOREIGN KEY ("actorId") REFERENCES
"users"("id") ON DELETE SET NULL ON UPDATE CASCADE;

-- Migration: 20260424_phase12b_hardening_indexes
-- Phase 12B hardening indexes
-- Speeds up buyer order listings and payment gateway/idempotency event lookups.

CREATE INDEX IF NOT EXISTS "orders_buyerId_createdAt_idx"
ON "orders"("buyerId", "createdAt");

CREATE INDEX IF NOT EXISTS "payments_gatewayRef_idx"
ON "payments"("gatewayRef");

CREATE INDEX IF NOT EXISTS "payment_events_createdAt_idx"
ON "payment_events"("createdAt");

-- Migration: 20260503_add_buyer_business_type
-- CreateEnum
CREATE TYPE "BuyerBusinessType" AS ENUM ('INDIVIDUAL', 'RESTAURANT', 'HOTEL', 'WHOLESALE', 'RETAIL',
'OTHER');

-- AlterTable
ALTER TABLE "buyer_profiles" ADD COLUMN "businessType" "BuyerBusinessType";

-- Migration: 20260503_add_producer_coordinates
-- AlterTable
ALTER TABLE "producer_profiles"
ADD COLUMN "latitude" DECIMAL(9,6),
ADD COLUMN "longitude" DECIMAL(9,6);

-- Migration: 20260503_product_categories_foundation
-- AlterTable: categories multilingual fields
ALTER TABLE "categories" ADD COLUMN "nameEn" TEXT;
ALTER TABLE "categories" ADD COLUMN "nameAr" TEXT;

UPDATE "categories"
SET
"nameEn" = COALESCE("name"->>'en', "slug"),
"nameAr" = COALESCE("name"->>'ar', "slug");

ALTER TABLE "categories" ALTER COLUMN "nameEn" SET NOT NULL;
ALTER TABLE "categories" ALTER COLUMN "nameAr" SET NOT NULL;

ALTER TABLE "categories" DROP COLUMN "name";

-- Migration: 20260504_add_favorites
CREATE TABLE "favorite_products" (
"id" TEXT NOT NULL,
"buyerId" TEXT NOT NULL,
"productId" TEXT NOT NULL,
"createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
CONSTRAINT "favorite_products_pkey" PRIMARY KEY ("id")
);
```

```

CREATE TABLE "favorite_producers" (
  "id" TEXT NOT NULL,
  "buyerId" TEXT NOT NULL,
  "producerId" TEXT NOT NULL,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT "favorite_producers_pkey" PRIMARY KEY ("id")
);

CREATE UNIQUE INDEX "favorite_products_buyerId_productId_key" ON "favorite_products"("buyerId", "productId");
CREATE INDEX "favorite_products_buyerId_idx" ON "favorite_products"("buyerId");
CREATE INDEX "favorite_products_productId_idx" ON "favorite_products"("productId");

CREATE UNIQUE INDEX "favorite_producers_buyerId_producerId_key" ON "favorite_producers"("buyerId",
  "producerId");
CREATE INDEX "favorite_producers_buyerId_idx" ON "favorite_producers"("buyerId");
CREATE INDEX "favorite_producers_producerId_idx" ON "favorite_producers"("producerId");

ALTER TABLE "favorite_products" ADD CONSTRAINT "favorite_products_buyerId_fkey" FOREIGN KEY ("buyerId")
  REFERENCES "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE "favorite_products" ADD CONSTRAINT "favorite_products_productId_fkey" FOREIGN KEY ("productId")
  REFERENCES "products"("id") ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE "favorite_producers" ADD CONSTRAINT "favorite_producers_buyerId_fkey" FOREIGN KEY ("buyerId")
  REFERENCES "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE "favorite_producers" ADD CONSTRAINT "favorite_producers_producerId_fkey" FOREIGN KEY
  ("producerId") REFERENCES "producer_profiles"("id") ON DELETE CASCADE ON UPDATE CASCADE;

-- Migration: 20260504_add_producer_nis
-- AlterTable
ALTER TABLE "producer_profiles" ADD COLUMN "nis" TEXT;

-- CreateIndex
CREATE UNIQUE INDEX "producer_profiles_nis_key" ON "producer_profiles"("nis");

-- Migration: 20260504_add_product_recipe_pdf
-- AlterTable
ALTER TABLE "products" ADD COLUMN "recipePdfUrl" TEXT;

-- Migration: 20260504_add_reviews
CREATE TABLE "reviews" (
  "id" TEXT NOT NULL,
  "orderId" TEXT NOT NULL,
  "productId" TEXT NOT NULL,
  "producerId" TEXT NOT NULL,
  "buyerId" TEXT NOT NULL,
  "rating" INTEGER NOT NULL,
  "comment" TEXT,
  "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT "reviews_pkey" PRIMARY KEY ("id"),
  CONSTRAINT "reviews_rating_check" CHECK ("rating" >= 1 AND "rating" <= 5)
);

CREATE UNIQUE INDEX "reviews_buyerId_productId_orderId_key" ON "reviews"("buyerId", "productId", "orderId");
CREATE INDEX "reviews_productId_idx" ON "reviews"("productId");
CREATE INDEX "reviews_producerId_idx" ON "reviews"("producerId");
CREATE INDEX "reviews_buyerId_idx" ON "reviews"("buyerId");
CREATE INDEX "reviews_orderId_idx" ON "reviews"("orderId");

ALTER TABLE "reviews" ADD CONSTRAINT "reviews_orderId_fkey" FOREIGN KEY ("orderId") REFERENCES "orders"("id")
  ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE "reviews" ADD CONSTRAINT "reviews_productId_fkey" FOREIGN KEY ("productId") REFERENCES
  "products"("id") ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE "reviews" ADD CONSTRAINT "reviews_producerId_fkey" FOREIGN KEY ("producerId") REFERENCES
  "producer_profiles"("id") ON DELETE CASCADE ON UPDATE CASCADE;
ALTER TABLE "reviews" ADD CONSTRAINT "reviews_buyerId_fkey" FOREIGN KEY ("buyerId") REFERENCES "users"("id")
  ON DELETE CASCADE ON UPDATE CASCADE;

-- Migration: 20260504_coupons_mvp
CREATE TYPE "CouponType" AS ENUM ('PERCENT', 'FIXED');

CREATE TABLE "coupons" (

```

FarmConnect Database Schema (PostgreSQL)

```
"id" TEXT NOT NULL,
"code" TEXT NOT NULL,
"producerId" TEXT NOT NULL,
"type" "CouponType" NOT NULL,
"amount" DECIMAL(10,2) NOT NULL,
"startsAt" TIMESTAMP(3) NOT NULL,
"endsAt" TIMESTAMP(3) NOT NULL,
"usageLimit" INTEGER,
"usedCount" INTEGER NOT NULL DEFAULT 0,
"isActive" BOOLEAN NOT NULL DEFAULT true,
"createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,
"updatedAt" TIMESTAMP(3) NOT NULL,
CONSTRAINT "coupons_pkey" PRIMARY KEY ("id")
);

CREATE UNIQUE INDEX "coupons_code_key" ON "coupons"("code");
CREATE INDEX "coupons_producerId_idx" ON "coupons"("producerId");
CREATE INDEX "coupons_code_idx" ON "coupons"("code");

ALTER TABLE "coupons" ADD CONSTRAINT "coupons_producerId_fkey" FOREIGN KEY ("producerId") REFERENCES
    "producer_profiles"("id") ON DELETE CASCADE ON UPDATE CASCADE;

ALTER TABLE "orders"
ADD COLUMN "couponCode" TEXT,
ADD COLUMN "discountAmount" DECIMAL(10,2);

-- Migration: 20260504_delivery_option_mvp
CREATE TYPE "DeliveryMethod" AS ENUM ('PICKUP', 'DELIVERY');

ALTER TABLE "producer_profiles"
ADD COLUMN "producerOffersDelivery" BOOLEAN NOT NULL DEFAULT false;

ALTER TABLE "orders"
ADD COLUMN "deliveryMethod" "DeliveryMethod" NOT NULL DEFAULT 'PICKUP',
ADD COLUMN "deliveryFee" DECIMAL(10,2);

-- Migration: 20260504_delivery_qr_verification
-- AlterTable
ALTER TABLE "orders"
ADD COLUMN "deliveryVerificationToken" TEXT,
ADD COLUMN "verifiedAt" TIMESTAMP(3);

-- Migration: 20260504_harvest_logs_mvp
-- CreateEnum
CREATE TYPE "ProductUnit" AS ENUM ('KG', 'PIECE', 'BOX');

-- CreateEnum
CREATE TYPE "ProductLogType" AS ENUM ('WATERING', 'HARVEST', 'FERTILIZE', 'OTHER');

-- AlterTable
ALTER TABLE "products"
ADD COLUMN "harvestDate" TIMESTAMP(3),
ADD COLUMN "harvestWindowStart" TIMESTAMP(3),
ADD COLUMN "harvestWindowEnd" TIMESTAMP(3),
ADD COLUMN "isSeasonal" BOOLEAN NOT NULL DEFAULT false,
ADD COLUMN "seasonStartMonth" INTEGER,
ADD COLUMN "seasonEndMonth" INTEGER;

-- Add unit enum column with cast from existing values
ALTER TABLE "products" ADD COLUMN "unit_new" "ProductUnit" NOT NULL DEFAULT 'KG';

UPDATE "products"
SET "unit_new" = CASE
    WHEN lower("unit") = 'piece' THEN 'PIECE'::"ProductUnit"
    WHEN lower("unit") = 'box' THEN 'BOX'::"ProductUnit"
    ELSE 'KG'::"ProductUnit"
END;

ALTER TABLE "products" DROP COLUMN "unit";
ALTER TABLE "products" RENAME COLUMN "unit_new" TO "unit";

-- CreateTable
```

FarmConnect Database Schema (PostgreSQL)

```
CREATE TABLE "product_logs" (  
    "id" TEXT NOT NULL,  
    "productId" TEXT NOT NULL,  
    "type" "ProductLogType" NOT NULL,  
    "note" TEXT NOT NULL,  
    "happenedAt" TIMESTAMP(3) NOT NULL,  
    "createdBy" TEXT NOT NULL,  
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    CONSTRAINT "product_logs_pkey" PRIMARY KEY ("id")  
);  
  
-- CreateIndex  
CREATE INDEX "product_logs_productId_happenedAt_idx" ON "product_logs"("productId", "happenedAt");  
  
-- AddForeignKey  
ALTER TABLE "product_logs" ADD CONSTRAINT "product_logs_productId_fkey" FOREIGN KEY ("productId") REFERENCES  
    "products"("id") ON DELETE CASCADE ON UPDATE CASCADE;  
  
-- Migration: 20260504_reports_mvp  
-- CreateEnum  
CREATE TYPE "ReportTargetType" AS ENUM ('USER', 'PRODUCT', 'ORDER');  
  
-- CreateEnum  
CREATE TYPE "ReportReason" AS ENUM ('SPAM', 'FRAUD', 'ABUSE', 'INAPPROPRIATE_CONTENT', 'OTHER');  
  
-- CreateEnum  
CREATE TYPE "ReportStatus" AS ENUM ('OPEN', 'REVIEWING', 'RESOLVED', 'REJECTED');  
  
-- CreateTable  
CREATE TABLE "reports" (  
    "id" TEXT NOT NULL,  
    "reporterId" TEXT NOT NULL,  
    "targetType" "ReportTargetType" NOT NULL,  
    "targetId" TEXT NOT NULL,  
    "reason" "ReportReason" NOT NULL,  
    "description" TEXT NOT NULL,  
    "status" "ReportStatus" NOT NULL DEFAULT 'OPEN',  
    "internalNote" TEXT,  
    "createdAt" TIMESTAMP(3) NOT NULL DEFAULT CURRENT_TIMESTAMP,  
    "updatedAt" TIMESTAMP(3) NOT NULL,  
  
    CONSTRAINT "reports_pkey" PRIMARY KEY ("id")  
);  
  
-- CreateIndex  
CREATE INDEX "reports_reporterId_createdAt_idx" ON "reports"("reporterId", "createdAt");  
CREATE INDEX "reports_targetType_targetId_idx" ON "reports"("targetType", "targetId");  
CREATE INDEX "reports_status_createdAt_idx" ON "reports"("status", "createdAt");  
  
-- AddForeignKey  
ALTER TABLE "reports" ADD CONSTRAINT "reports_reporterId_fkey" FOREIGN KEY ("reporterId") REFERENCES  
    "users"("id") ON DELETE CASCADE ON UPDATE CASCADE;
```